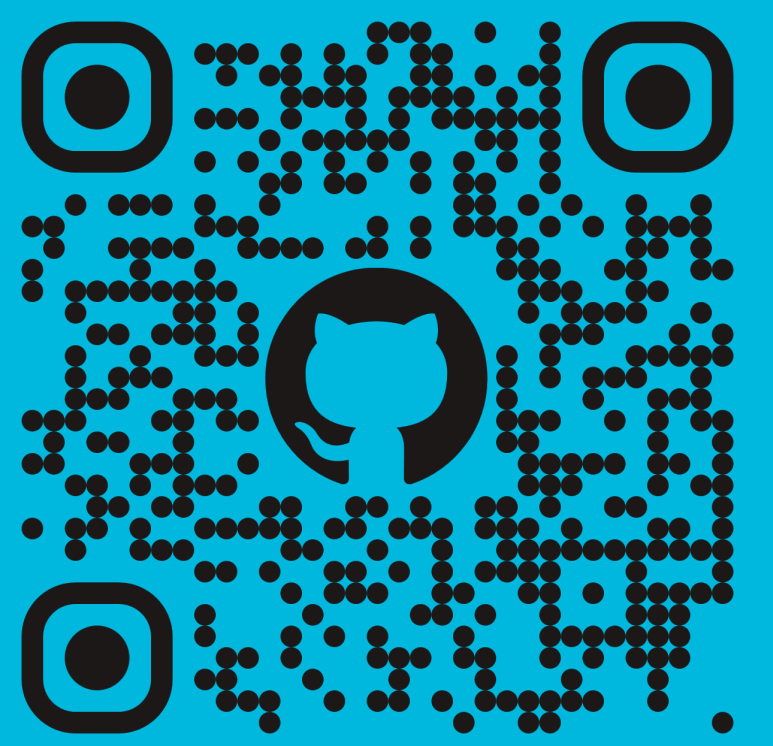




Poster Summary

We introduce **AutoMashup**, an open-source system for automatic mashup creation using deep learning tools (**Demucs** [1] and **All-in-1** [2]).

We propose using the **COCOLA** [3] model to assess musical compatibility and investigate if general-purpose audio models (**CLAP** [4], **MERT** [5]) can estimate this compatibility in a zero-shot setting to overcome COCOLA's computational cost.

Our key findings reveal that mashup compatibility is fundamentally **asymmetric**, and that current general-purpose embeddings **fail to capture** the musical coherence required for this task.

All code, experiments, some mashups, and the AutoMashup tool are publicly available on GitHub: <https://github.com/ax-le/automashup>

The AutoMashup Pipeline

AutoMashup aligns two songs harmonically and rhythmically to create a coherent mashup. The process involves source separation, music analysis, and alignment.

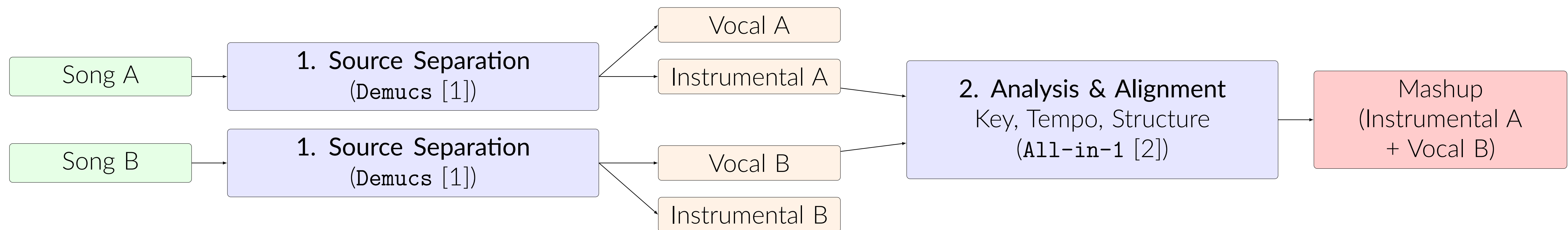


Figure 1. The AutoMashup workflow, from input songs to final mashup.

The Song Selection Problem

Manually finding compatible tracks is difficult. We need an automatic method to predict mashup quality.

- **Goal:** Predict compatibility between a vocal stem and an instrumental stem.
- **1st solution:** **COCOLA** [3], a model trained for musical coherence, provides a qualitatively reliable score.
- **Problem:** COCOLA is computationally expensive ($O(n^2)$ complexity for a library of n songs), as it requires pairwise audio comparison.
- **Hypothesis:** Can we use pre-trained embeddings from large audio models (**CLAP** [4], **MERT** [5]) for a scalable $O(n)$ search?
→ We test if the distance between embeddings correlates with the COCOLA score.

Finding 1: Compatibility is Asymmetric

A key finding is that mashup compatibility is **not a symmetric relationship**. Defining the score as the cosine similarity in CLAP and MERT latent spaces, we found that:

$$\text{Score}(\text{Vocal A \& Inst B}) \neq \text{Score}(\text{Vocal B \& Inst A})$$

Hence, the quality of a mashup depends on the functional role (vocal vs. accompaniment) assigned to each track.

This implies that **compatibility cannot be derived from symmetric musical features** (like tonality or tempo) alone, a crucial consideration for modeling that was not explicit in prior work.

Finding 2: Latent Space Structure

We visualize the embeddings of 21 songs (vocal & instrumental) using t-SNE to understand their structure.

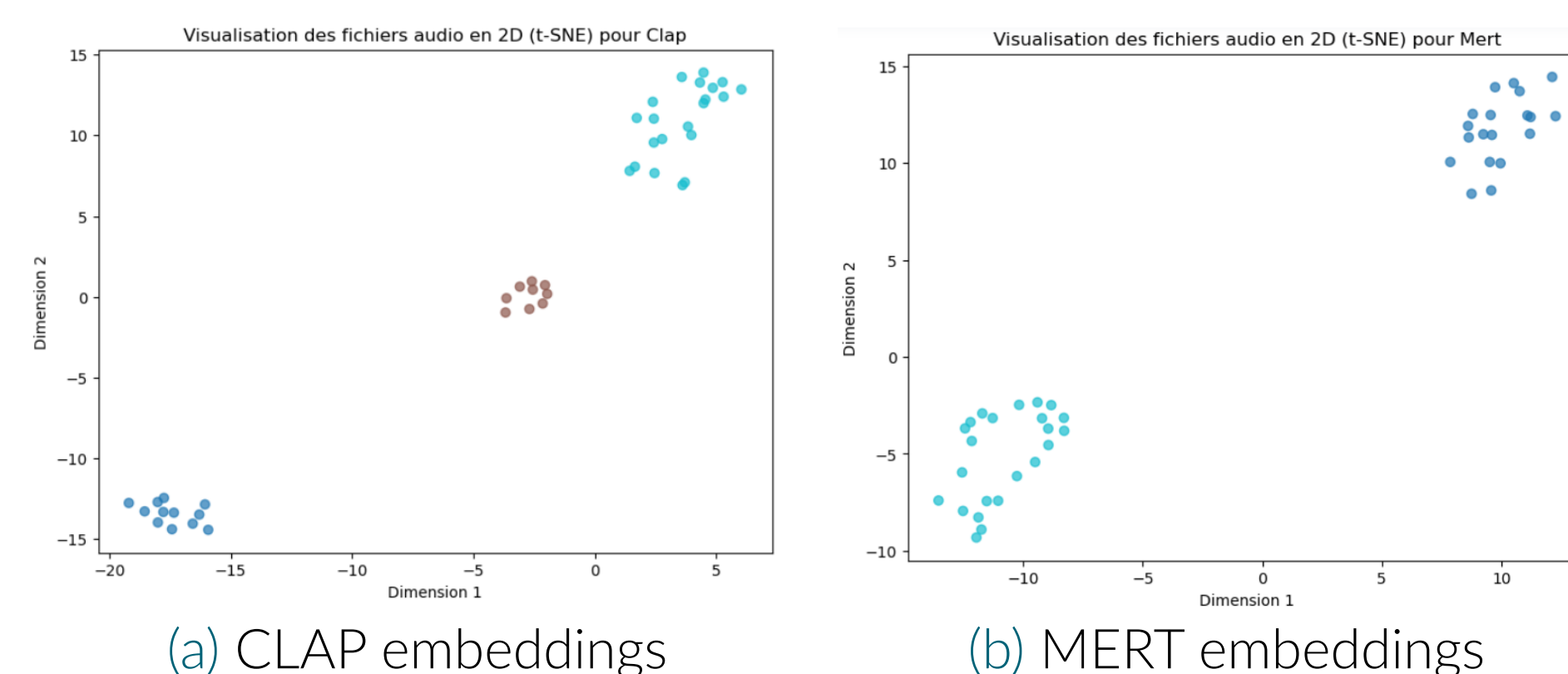


Figure 2. t-SNE visualization of vocal and instrumental embeddings. Colors indicate different clusters. A modality gap appears in CLAP.

Clusterings are dissimilar (Adjusted Rand Index = -0.026), indicating that models learn different representations.

Finding 3: No Correlation with COCOLA

Focusing on 21 songs and all their potential pairs (Instrumental A + Vocal B), we computed the correlation between:

- The cosine similarity of embeddings (CLAP or MERT),
- The COCOLA compatibility score.

Correlation Metric	CLAP	MERT
Pearson Correlation	0.051	-0.018
Spearman Rank Corr.	0.079	-0.017
Kendall's Tau Corr.	0.053	-0.010

Conclusion: There is **no meaningful correlation**.

The similarity of tested general-purpose embeddings is not predictive of musical compatibility for mashups in a zero-shot setting.

Take-Home Messages

1. **A New Open-Source Tool:** AutoMashup provides a full, reproducible pipeline for creating music mashups. <https://github.com/ax-le/automashup>
2. **Compatibility is Asymmetric:** The functional role of each track (vocal vs. instrumental) is crucial for compatibility, and was previously overlooked.
3. **Embeddings Fall Short:** CLAP and MERT (state-of-the-art foundation audio models) are not suited for zero-shot compatibility estimation in this creative context.

Perspectives & Future Work

1. **Fine-tuning / Distillation:** Use COCOLA scores to adapt large audio models to the compatibility task.
2. **Role-Aware Representations:** Develop compatibility estimation models that explicitly account for a stem's functional role.
3. **Human-in-the-Loop:** Conduct perceptual studies to validate automatic metrics and guide model development.

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